

L2 - 14.3. Osquery The Basics

Task 1 - Introduction

Osquery is an open-source, multi-platform monitoring agent created by Facebook that allows security professionals to query operating system data as a relational database using SQL.

Task 2 - Connect with the Lab

The lab machine attached to this room already has Osquery installed and configured for you on Windows and Linux. Click on the `powershell terminal` pinned at the taskbar and enter `osqueryi` to enter the interactive mode of osquery.

Task 3 - Osquery: Interactive Mode

One of the ways to interact with Osquery is by using the interactive mode. Open the terminal and run `osqueryi`.

```
root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery> .help
```

To list all the available tables that can be queried, use the `.tables` meta-command.

```
root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery> .table
=> appcompat_shims
=> arp_cache
=> atom_packages
=> authenticcode
```

To list all the tables with the term `user` in them, we will use `.tables user` as shown below:

```

root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery> .table user
=> user_groups
=> user_ssh_keys

```

Understanding the table Schema: We can list a table's schema with the following meta-command: `.schema table_name`

```

root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery> .schema users
CREATE TABLE users(`uid` BIGINT, `gid` BIGINT

```

SQL QUERY SYNTAX: `select column1, column2, column3 from table;`

```

root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery> select gid, uid, description, username, directory from
m users;

```

Display Mode: Osquery comes with multiple display modes to select from.

```

root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'

```

Answer the questions below

How many tables are returned when we query "table process" in the interactive mode of Osquery?

3

```

osquery> .tables process
=> process_memory_map
=> process_open_sockets
=> processes

```

Looking at the schema of the processes table, which column displays the process id for the particular process?

pid

Examine the .help command, how many output display modes are available for the .mode command?

5

```
Set output mode where MODE is one of:
csv      Comma-separated values
column   Left-aligned columns see .width
line     One value per line
list     Values delimited by .separator string
pretty   Pretty printed SQL results (default)
```

Task 4 - Schema Documentation

106 Tables

Osquery Version: 5.5.1 (current)

Show only Tables compatible with: Windows

appcompat_shims

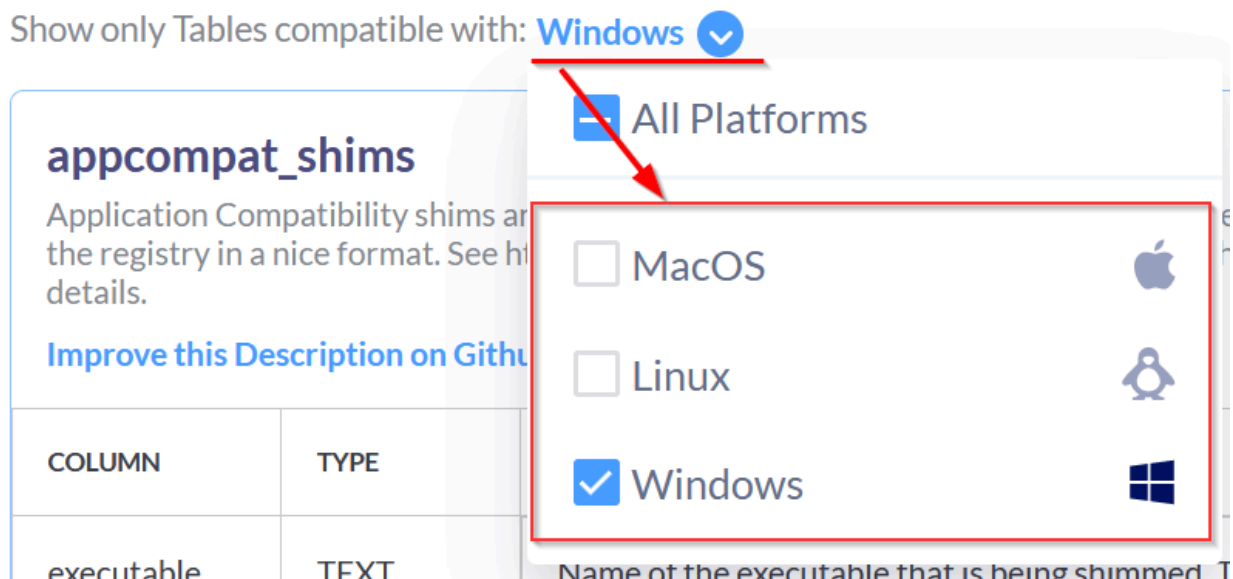
Application Compatibility shims are a way to persist malware. This table presents the AppCompat Shim information from the registry in a nice format. See http://files.brucon.org/2015/Tomczak_and_Ballenthin_Shims_for_the_Win.pdf for more details.

[Improve this Description on Github](#)

COLUMN	TYPE	DESCRIPTION
executable	TEXT	Name of the executable that is being shimmed. This is pulled from the registry.
path	TEXT	This is the path to the SDB database.
description	TEXT	Description of the SDB.
install_time	INTEGER	Install time of the SDB
type	TEXT	Type of the SDB database.
sdb_id	TEXT	Unique GUID of the SDB.

1. A dropdown lists various versions of Osquery. Choose the version of Osquery you wish to see schema tables for.
2. The number of tables within the selected version of Osquery. (In the above image, 106 tables are available).

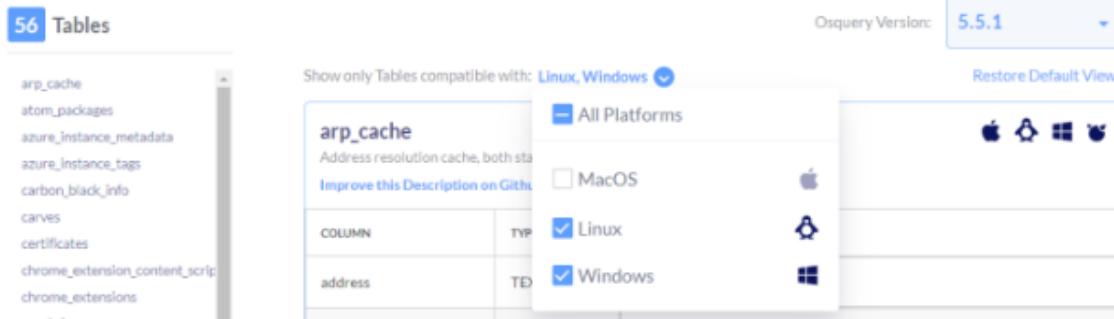
3. The list of tables is listed in alphabetical order for the selected version of Osquery. This is the same result we get when we use the `.table` command in the interactive mode.
4. The name of the table and a brief description.
5. A detailed chart showing each table's **column**, **type**, and **description**.
6. Information to which Operating System the table applies. (In the above image, the **account_policy_data** table is available only for **macOS**)
7. A dropdown menu to select the Operating System of choice. We can choose multiple Operating Systems, which will display the tables available for those Operating systems.



Answer the questions below

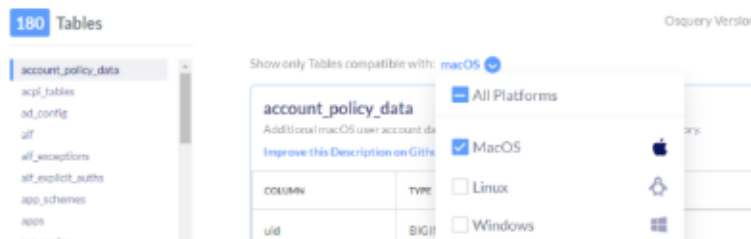
In Osquery version 5.5.1, how many common tables are returned, when we select both Linux and Window Operating system?

56



In Osquery version 5.5.1, how many tables for MAC OS are available?

180



In the Windows Operating system, which table is used to display the installed programs?

programs

In Windows Operating system, which column contains the registry value within the registry table?

data

Task 5 - Creating SQL Queries

The SQL language implemented in Osquery is not an entire SQL language that you might be accustomed to, but rather it's a superset of SQLite. Realistically all your queries will start with a **SELECT** statement. This makes sense because, with Osquery, you are only querying information on an endpoint. You won't be updating or deleting any information/data on the endpoint.

Exploring Installed Programs: If you wish to retrieve all the information about the installed programs on the endpoint, first understand the table schema either using

the `.schema programs` command in the interactive mode.

```
root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery>select * from programs limit 1;
```

In the above example `LIMIT` was used followed by the number to limit the results to display.

Query : `SELECT name, version, install_location, install_date from programs limit 1;`

```
root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery>select name, version, install_location, install_date
from programs limit 1;
```

The above query will list the name, version, install location, and installed date of the programs on the endpoint. This will still return many results, depending on how busy the endpoint is.

Count: To see how many programs or entries in any table are returned, we can use the **count()** function, as shown below:

Query : `SELECT count(*) from programs;`

```
root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery>select count(*) from programs;
count(*) = 160
```

WHERE Clause uses to narrow down the list of results returned based on specified criteria. The following query will first get the user table and only display the result for the user James, as shown below:

Query : `SELECT * FROM users WHERE username='James';`

```
root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
```

```
osquery>SELECT * FROM users WHERE username='James';
      uid = 1002
      gid = 544
```

The equal sign is not the only filtering option in a **WHERE** clause. Below are filtering operators that can be used in a **WHERE** clause:

- `=` [equal]
- `<>` [not equal]
- `>` , `>=` [greater than, greater than, or equal to]
- `<` , `<=` [less than or less than or equal to]
- `BETWEEN` [between a range]
- `LIKE` [pattern wildcard searches]
- `%` [wildcard, multiple characters]
- `_` [wildcard, one character]

Matching Wildcard Rules

- `%` : Match all files and folders for one level.
- `%%` : Match all files and folders recursively.
- `%abc` : Match all within-level ending in "abc".
- `abc%` : Match all within-level starting with "abc".

Matching Examples

- `/Users/%/Library` : Monitor for changes to every user's Library folder, *but not the contents within* .
- `/Users/%/Library/` : Monitor for changes to files *within* each Library folder, but not the contents of their subdirectories.
- `/Users/%/Library/%` : Same, changes to files within each Library folder.
- `/Users/%/Library/%%` : Monitor changes recursively within each Library.
- `/bin/%sh` : Monitor the `bin` directory for changes ending in `sh` .

Joining Tables using JOIN Function: OSquery can also be used to join two tables based on a column that is shared by both tables.

Query1: `select uid, pid, name, path from processes;`

Query2: `select uid, username, description from users;`

Joined Query: `select p.pid, p.name, p.path, u.username from processes p JOIN users u on u.uid=p.uid LIMIT 10;`

```
root@analyst$ osqueryi
Using a virtual database. Need help, type '.help'
osquery>select p.pid, p.name, p.path, u.username from processes p JOIN users u on u.uid=p.uid LIMIT 10;
+-----+-----+-----+-----+
-----+-----+
| pid   | name                | path                                |
| username |
+-----+-----+-----+-----+
-----+-----+
| 7560   | sihost.exe          | C:\Windows\System32\sihost.exe     |
| James  |                      |                                     |
| 6984   | svchost.exe         | C:\Windows\System32\svchost.exe    |
| James  |                      |                                     |
| 7100   | svchost.exe         | C:\Windows\System32\svchost.exe    |
| James  |                      |
```

Answer the questions below

| Using Osquery, how many programs are installed on this host?

19

```
osquery> select count (*) from programs;
+-----+
| count (*) |
+-----+
| 19        |
+-----+
```

| Using Osquery, what is the description for the user James?

Creative Artist

When we run the following search query, what is the full SID of the user with RID '1009'?

S-1-5-21-1966530601-3185510712-10604624-1009

```
Query: select path, key, name from registry where key = 'HKEY_USERS';
```

S-1-5-21-1966530601-3185510712-10604624-1009

When we run the following search query, what is the Internet Explorer browser extension installed on this machine?

C:\Windows\System32\ieframe.dll

```
Query: select * from ie_extensions;  
C:\Windows\System32\ieframe.dll
```

After running the following query, what is the full name of the program returned?

```
Query: select name,install_location from programs where name  
LIKE '%wireshark%';
```

Wireshark 4.4.9 x64

Task 6 - Challenge and Conclusion

Now that we have explored various tables, learned how to create search queries, and ask questions from the operating system, it's time for a challenge. Use OSquery to examine the host and answer the following questions.

Answer the questions below

Which table stores the evidence of process execution in Windows OS?

userassist

One of the users seems to have executed a program to remove traces from the disk; what is the name of that program?

DiskWipe.exe

Create a search query to identify the VPN installed on this host. What is name of the software?

ProtonVPN

```
select name,install_location from programs where name like '%VPN%';
```

How many services are running on this host?

215

```
select count(name) from services;
```

A table autoexec contains the list of executables that are automatically executed on the lab machine. There seems to be a batch file that runs automatically. What is the name of that batch file (with the extension .bat)?

batstartup.bat

```
select name from autoexec where name like '%.bat';
```

What is the full path of the batch file found in the above question? (Last in the List)

C:\Users\James\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\batstartup.bat

```
select name,path from autoexec where name like '%.bat';
```